Cost of one relation trial on a high-genus binary Jacobian

(2026-10-24) Prices the open unit conversion in `ghs_counterfactual_n130_20261013` finding 6. Frozen design: `PROTOCOL.md` (written before any data). Labels: **measured** (counted by `trial_cost.rs`), **derived** (arithmetic on measured values), **model** (the note's cost model, or a fit extrapolation), **assumption**. This is a model-pricing study of an abstract tower; it attacks no registry curve and builds no GHS cover.

Result in one paragraph

One relation trial (one Cantor addition on `y^2+hy=f` over F_2, then one distinct-degree smoothness test of `u(x)` at the model's `b`) costs **198,149 word operations at g = 255** (measured, `b = 27`), which is **103 additions of a point on a binary curve over F_2^131 = 2^6.69** (derived, same counted unit; the note guessed 2^7). Substituting it into the N = 130 model moves the margin from +4.71 bits to **−1.83 bits** (derived): the index-calculus cost becomes 2^62.15 against rho 2^60.31. The note's own 2^7 gives −2.15. The margin survives only while one trial costs at most about 29 curve additions (2^4.85, derived); the smoothness test alone costs 16.5 (2^4.04), so no Cantor step, however fast, leaves more than 0.81 bit.

What a trial is (read from `window.rs::price`)

`price` charges `relations = log2|FB| - log2 P(smooth)`, `linear algebra = 2 log2|FB| + log2 g`, `total = max + 1`; its doc says one operation is "one divisor-class step plus one smoothness test". The trial measured: `D <- D + A_r` (Cantor, `A_r` from a 16-entry table of random classes, one group step per trial), then smooth(`u(D)`, `b`) by distinct-degree factorisation with the early exits of PROTOCOL section

2. Unit **W** = executed F_2[x] u64 word operations (carry-less

shift-xor, division, squaring), identical code for the F_2^131 curve addition used as the denominator. `b` per genus is the model's optimum (f64 port of `price` reproduces the note: g = 128 b 17 -> 36.41, g = 255 b 27 -> 55.60, g = 256 -> 55.65; control passed).

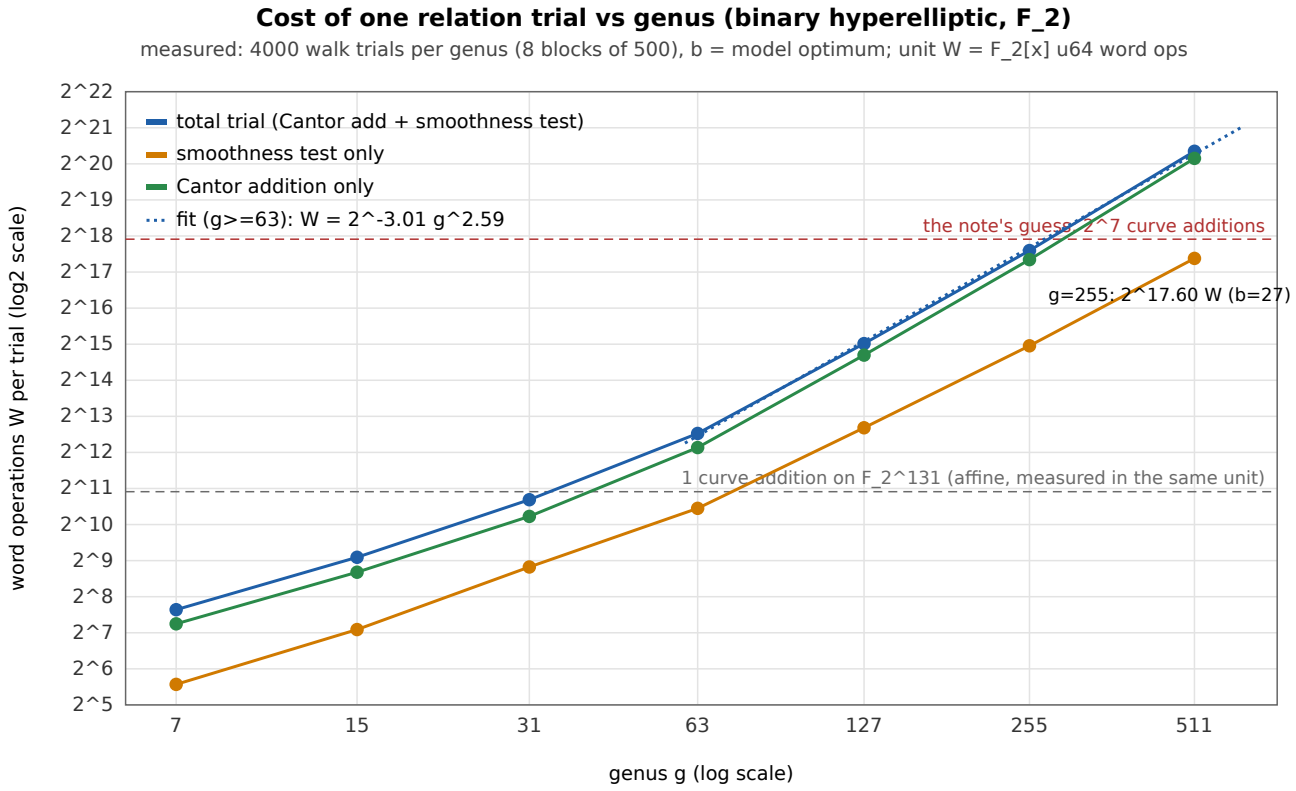
Cost per trial against genus (measured; 4000 trials per genus, 8

blocks of 500)

g	b (model)	W Cantor add	of which compose	W smooth test	**W / trial**	block sd	native events	reduction steps	curve additions (W / 1925)
7	2	152	108	47	**199**	2	30.5	1.8	0.10
15	3	410	243	136	**546**	6	45.2	3.6	0.28
31	6	1,197	533	453	**1,650**	13	69.8	7.5	0.86
63	10	4,497	730	1,398	**5,895**	37	106.8	15.8	3.06
127	17	26,562	2,890	6,575	**33,137**	137	198.3	31.6	17.2
255	27	166,385	8,622	31,764	**198,149**	940	357.3	63.6	**102.9 (2^6.69)**
511	42	1,164,425	32,255	170,378	**1,334,802**	3,093	660.5	127.5	693 (2^9.44)

Reference (measured, W per op on F_2^131, `z^131+z^8+z^3+z^2+1`): multiply 501, square 142, inversion (Euclid) 761, **affine curve addition 1,925**. The Lopez-Dahab mixed addition priced as 8M+5S+9A is 4,741 (model op count), so the affine addition is the smaller denominator and is the headline one (PROTOCOL section 3: smaller denominator, conservative against the margin). Wall time per trial (practicality only, shared host): g = 255, about 0.42 ms; affine addition 9.1 us; ratio 46, not used.

Graph:



(`cost_per_trial_vs_genus.svg` / `.png`, drawn by `trial_cost.rs` from `results.json`; the PNG is a 150 dpi render of the SVG).

Scaling (fit of log2 W on log2 g; measured points, fit is model)

fit range	total	Cantor add	smoothness test
g = 63..511	W = 2^-3.01 g^2.59	slope 2.65	slope 2.29
g = 15..511	W = 2^-0.23 g^2.23	slope 2.28	slope 2.02

Extrapolation to g = 255 from the g = 63..511 fit: 2^17.68 W (model) against the direct measurement 2^17.60 (0.08 bit); from g = 31, 63, 127 only: 2^17.04 (0.56 bit low). Direct measurement at 255 is what the margin below uses.

Preregistered predictions against what happened

prediction (PROTOCOL section 6)	outcome
Cantor add slope in [1.8, 2.3] over 63..511	**missed**: 2.65
total slope in [2.2, 2.8]	hit: 2.59
test slope in [2.2, 2.8] (stated for the total)	2.29
smoothness test dominates the Cantor add for g >= 63	**wrong**: at g = 255 the add is 84% of the trial
EC addition in [2^11, 2^13] W	affine 2^10.91 (just below); LD model 2^12.2 (inside)
CAE at g = 255 in [2^5, 2^9]	hit: 2^6.69

Why the Cantor slope missed: the textbook reduction loop runs about g/4 steps (measured 63.6 at g = 255), each with full-size products and a division, so the reduction costs 158k of the 166k W while composition is 8.6k. A Cantor-type algorithm with an O(g^2) reduction (interleaved composition and reduction, Harley or NUCOMP style) is not implemented here; scenarios B and C below bracket it.

N = 130 margin with the measured conversion (g = 255, rho 2^60.315)

Model terms at `b = 27`: `rel = 54.462`, `lin = 54.603`, total `2^55.603`, margin +4.711 (the note). The conversion enters the relation term: `total = max(rel + log2 CAE, lin) + 1`. Because `lin - rel = 0.14` bit, the margin is `4.711 - (log2 CAE - 0.14)` once the conversion exceeds 0.14 bit.

scenario	W / trial	curve additions	total	margin vs rho	label
note's guess	-	2^7	2^62.46	−2.15	model
A: measured trial (textbook Cantor + test)	198,149	2^6.69 (103)	2^62.15	**−1.83**	measured/derived

B: composition only + test (no reduction cost)	40,387	2^4.39	2^59.85	+0.46	lower bound for a Cantor-type step
C: smoothness test only (divisor step free)	31,764	2^4.04	2^59.51	+0.81	floor under this test

Re-optimising `b` with the measured cost per `b` (scan b = 8..64 at g = 255, 300 trials each, `results.json` `bscan255`) keeps `b = 27` best, margin −1.83 (derived). The margin does not survive the measured conversion.

Section 5a band (unpriced native work at 0, 0.1, 1 GAE per event)

Native event = one polynomial primitive call (mul, sqr, divmod, gcd/xgcd), counted in the trial (A: 357.3, B: 102.7, C: 90.7 events). IC trial = 1 divisor-class step + w x events; rho step = 1 + w x n_rho with **assumed** n_rho = 3 native events per step (canonicalisation, distinguished-point test, partition hash; not measured here), sensitivity n_rho = 0 and 10. The linear-algebra term stays at the note's unit. Margin in bits (derived):

scenario	w = 0	w = 0.1	w = 1	sign across the band
A (n_rho 0 / 3 / 10)	+4.71	−0.35 / +0.03 / +0.65	−3.63 / −1.63 / −0.17	not positive at every weight: unresolved, negative at w = 1 and at the calibrated point
B (n_rho 0 / 3 / 10)	+4.71	+1.36 / +1.74 / +2.36	−1.84 / +0.16 / +1.61	positive only if n_rho >= 3 at w = 1: unresolved
C (n_rho 0 / 3 / 10)	+4.71	+1.52 / +1.90 / +2.52	−1.67 / +0.33 / +1.79	same

Only w = 0 reproduces the note's +4.71. Per section 5a the arms are not ranked: the "breaks rho: yes" entry of the N = 130 row is **unresolved at the band boundary** for the lower-bound scenarios and **fails** for the measured trial (A) at the calibrated point and at w = 1. The calibrated point (the ratio above) is shown beside the band, never in place of it.

Controls (all passed; `controls_failed = 0`)

- **Cantor outputs**, every g (7..511): 200 additions plus 16 table entries per g: `u` monic, `deg u <= g`, `deg v < deg u`, `u | v^2 + hv + f`; **0 failures**.
- **Group law**, every g: 100 triples each: associativity, commutativity, `A+0 = A`, `A + (-A) = 0`, `(A+A)+B = A+(A+B)`; **0 failures**. Curves are nonsingular (`gcd(h, h'^2 f + f'^2) = 1`). The order check `[ord]D = 0` was not run (the group order is not computed at these genera).
- **Smoothness vs full factorisation**: A, 300 random squarefree polynomials (deg 8..96, b 2..24) against DDF + equal-degree (trace map): 45 smooth, 255 not, **0 disagreements** B, 300 polynomials (deg <= 24, 295 with repeated factors, b 2..10) against trial division by all monic polynomials of degree <= 12: 248 smooth, 52 not, **0 disagreements**.
- **EC**: modulus irreducible; 60 points on curve; associativity on 50 triples; **0 failures**.
- **Model port**: matches the note at 128, 255, 256 to 0.01 bit and `b`.
- **Smooth-rate fidelity** (not gated; 200,000 trials each, one curve per g):

g	b	walk	random monic u	model P(smooth)
7	2	0.1474	0.1572	0.2745
11	2	0.00112	0.02105	0.0496
15	3	0.00170	0.01103	0.0243
23	4	0.00067	0.00093	0.00222
31	6	0.00221	0.00085	0.00160

The walk's smooth rate differs from the model's by factors from 0.02x (g = 11) to 1.4x (g = 31) and from random polynomials by up to 19x at small g (a single curve's class group is far from the random-polynomial picture at g <= 15, and the main sweep's g = 15 curve gave 4.6%, against 0.17% here). The model's divisor count includes the point at infinity (N_1 = 3), which a Mumford `u` omits, so the model is about 2x optimistic against random polynomials at g = 31. The effect on the g = 255 margin was not computed. The cost per trial does not depend on how often a trial succeeds, so it is unaffected; the relation count `1/P` is the model's.

Run record

field	value
source	`trial_cost.rs`, sha256 `b59da59417e9ef15f6a69f979984544b9efed92cc7806478657f9fe35acd58f5`
build and run	`rustc -C opt-level=3 -C target-cpu=native trial_cost.rs -o trial_cost && ./trial_cost <dir>` (about 50 s)
outputs	`results.json` sha256 `dc445045f6f8e97e0431f8d422f17369af90be0f9f4a62a96f269302a48df371`, `run.log`, SVG, PNG
seeds	master 20261024, per-g `splitmix64(master xor g)`

host	Linux 6.18.44, 4-core Intel Xeon (cloud VM), `rustc 1.97.0`, single thread, no isolation claimed; counts are deterministic (identical across the reruns), wall times and hence the JSON hash are not
failed or earlier runs	first run hung at g = 63 (a place-divisor sampler looped forever when the unique degree-2 irreducible had no solution of the quadratic); found with gdb, bounded the retries, reran from the start. A later edit added the composition/reduction split and the plot fixes; the reported run is the last one, all controls re-executed
PDF	`report.pdf` from `report.typ` (pandoc `-t typst` of this README, compiled with the `typst` 0.15.0 Python wheel in a scratch venv; no LaTeX engine is installed). Rendered and read

Not established

- Cost with a hardware carry-less multiply, an O(g^2) Cantor/NUCOMP step, a sieved or batched smoothness test, or a cheaper rho step (Koblitz Frobenius, bit-sliced); each moves the ratio. A rho step twice cheaper adds one bit to `log2 CAE`.
- Whether the walk (one Cantor step per trial) is the cheapest way to generate trials; a fresh random combination costs about k additions.
- The linear-algebra term in the model is kept at 1 operation per modular multiplication; it was not converted (it binds only within 0.14 bit).
- The rho reference is the note's `2^60.31`; no rho run was measured, and n_rho = 3 is an assumption.
- One curve per genus; no GHS cover, no Jacobian subgroup of an ECDLP instance, no relation to any registry curve.
- The Mumford-u smooth probability against the model (above).